



Lab 09: WLAN vs Wireless

WRT300N · Router0 (2911) · Switch0 · Printer · 3 PCs · Laptop

Lab #09

Topic: Wireless Networking

Tool: Cisco Packet Tracer

Cert: CompTIA Network+

1. Lab Objectives

- Build a combined wired LAN and wireless WLAN in one Packet Tracer topology.
- Configure Linksys WRT300N with SSID 'Yasin', WPA2-PSK and passphrase 'ahmed1234'.
- Connect Laptop0 wirelessly to the WRT300N on the 192.168.2.0/24 subnet.
- Connect PC0, PC1, PC2 and Printer0 to Switch0 on the wired 192.168.1.0/24 subnet.
- Route between the two subnets via Router0 (2911).

2. Network Topology

Switch0 (2960-24TT): Printer0 on port 0, PC0 on Fa0/1, PC1 on Fa0/2, PC2 on Fa0/3. Switch0 uplinks to Router0 (2911) on Fa0/4 → Gig0/0. Router0 Gig0/1 connects to WRT300N wireless router. Laptop0 connects wirelessly to WRT300N (SSID: Yasin, dashed line).

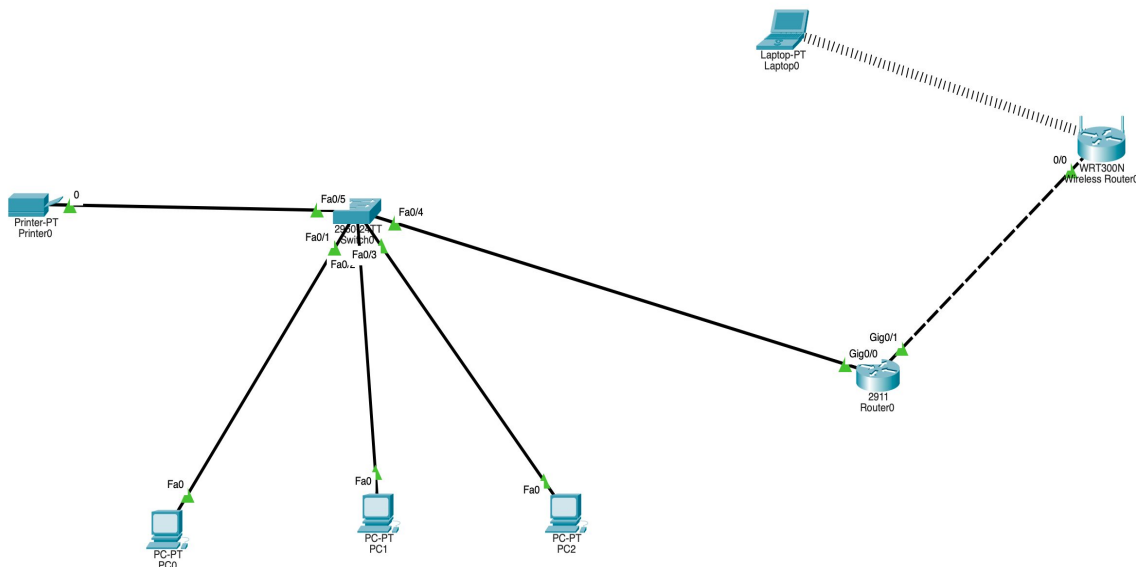


Figure 1 – Full topology: wired LAN (Switch0, PCs, Printer) + wireless (WRT300N, Laptop0)

3. WRT300N Configuration

3.1 Basic Setup – Network & DHCP



Wireless Settings

SSID: Yasin

2.4 GHz Channel: 1-2.412GHz

Coverage Range (meters): 250.00

Authentication:

- ☐ Disabled
- ☐ WEP
- ☒ WPA2-PSK
- ☐ WPA
- ☐ WPA2

WEP Key:

PSK Pass Phrase: ahmed1234

RADIUS Server Settings:

IP Address:

Shared Secret:

Encryption Type: AES

Figure 2 – WRT300N Basic Setup: Router IP 192.168.2.1, DHCP pool 192.168.2.100–149

Router IP	192.168.2.1
Subnet Mask	255.255.255.0
DHCP Server	Enabled
Start IP	192.168.2.100
Max Users	50
IP Range	192.168.2.100 – 192.168.2.149

3.2 Wireless Settings



Wireless-N Broadband Router

Firmware Version: v0.93.3

Setup | Setup | Wireless | Security | Access Restrictions | Applications & Gaming | Administration | Status

Basic Setup | DNS | MAC Address Clone | Advanced Routing

Internet Setup

Internet Connection type: Automatic Configuration - DHCP

Optional Settings (required by some internet service providers)

Host Name:

Domain Name:

MTU: Size: 1500

Network Setup

Router IP:

Subnet Mask: 255.255.255.0

DHCP Server: ☒ Enabled ☐ Disabled **DHCP Reservation**

Start IP Address: 192.168.2. 100

Maximum number of Users: 50

IP Address Range: 192.168.2. 100 - 149

Client Lease Time: minutes (0 means one day)

Static DNS 1: . . .

Static DNS 2: . . .

Static DNS 3: . . .

WINS: . . .

Figure 3 – Wireless tab: SSID 'Yasin', Mixed mode, Channel 1 (2.412 GHz), SSID Broadcast ON

SSID	Yasin
Network Mode	Mixed
Standard Channel	1 – 2.412 GHz
SSID Broadcast	Enabled

3.3 Wireless Security

Wireless-N Broadband Router

Firmware Version: v0.93.3

Wireless | Setup | Wireless | Security | Access Restrictions | Applications & Gaming | Administration | Status

Basic Wireless Settings | Wireless Security | Guest Network | Wireless MAC Filter | Advanced Wireless Settings

Basic Wireless Settings

Network Mode: Mixed

Network Name (SSID): Yasin

Radio Band: Auto

Wide Channel: Auto

Standard Channel: 1 - 2.412GHz

SSID Broadcast: ☒ Enabled ☐ Disabled

Figure 4 – Security: WPA2 Personal, AES encryption, Passphrase ahmed1234



Security Mode	WPA2 Personal
Encryption	AES
Passphrase	ahmed1234
Key Renewal	3600 seconds

3.4 Packet Tracer Wireless Interface Config

Figure 5 – PT wireless interface: SSID Yasin, WPA2-PSK, AES, PSK ahmed1234

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4. Router0 (2911) Configuration

```
Router0(config)#interface GigabitEthernet0/0
Router0(config-if)#ip address 192.168.1.1 255.255.255.0
Router0(config-if)#no shutdown
Router0(config-if)#exit
Router0(config)#interface GigabitEthernet0/1
Router0(config-if)#ip address 192.168.2.254 255.255.255.0
Router0(config-if)#no shutdown
Router0(config-if)#exit
Router0(config)#ip route 192.168.2.0 255.255.255.0 192.168.2.254
Router0#write memory
```

5. Verification

```
Laptop0>ping 192.168.2.1
Laptop0>ping 192.168.1.10
```



```
PC0>ping 192.168.2.100
```

```
Router0#show ip route
```

6. Key Takeaways

- The WRT300N is a home wireless router with built-in DHCP — different from an enterprise AP.
- SSID 'Yasin' is broadcast publicly so Laptop0 can discover and join the network.
- WPA2 Personal + AES is the correct secure setting for a small wireless network.
- Wired LAN and WLAN on different subnets require a router to communicate.
- The shared printer on the wired LAN is accessible from wireless clients only with routing.